Inference Server
Network Application
CAMARA Gateway
NEFSim
MQTT Broker

par [QoS monitoring and adaptation]

loop [on every KPI threshold evaluation]

1. POST {sink} CloudEvent
[type = org.camaraproject.connectivity-insights-subscriptions.v0.network-quality,
data (applicationProfileId, targetMinUpstreamRate, targetMinDownstreamRate)]

2. 204 No Content

opt ["unable to meet the application requirements" and no downgrade applied]

loop [for each pedestrian and vehicle UE]

3. POST /qod-provisioning/v0.2/retrieve-device-qos [device]

4. 200 OK [ProvisioningInfo (provisioningId)] | 404 Not Found

opt [UE provisioning exists]

5. DELETE /qod-provisioning/v0.2/device-qos/{provisioningId}

6. 204 No Content

7. POST /qod-provisioning/v0.2/device-qos
[device (ipv4Address | phoneNumber), qosProfile = downgrade profile, sink]

8. 201 Created [ProvisioningInfo (provisioningId, status = REQUESTED)]

9. POST {sink} CloudEvent
[type = org.camaraproject.qod-provisioning.v0.status-changed,
data (provisioningId, status = AVAILABLE | UNAVAILABLE)]

10. 204 No Content

[Pedestrian tracking]

loop [on every geofence transition]

11. POST {sink} CloudEvent
[type = org.camaraproject.geofencing-subscriptions.v0.area-entered | area-left,
data (device, area, subscriptionId)]

12. 204 No Content

13. Add or remove the pedestrian from the camera area set

[Collision detection]

loop [for each processed frame]

14. POST /callbacks/crash_inference/{cameraSupi}
[detections_number, predictions, frame_number]

15. 204 No Content

16. Filter by class and confidence threshold; confirm persistence across the frame window

opt [collision confirmed]

17. POST /location-retrieval/v0.4/retrieve [device (phoneNumber | ipv4Address)]

18. 200 OK [lastLocationTime, area (CIRCLE | POLYGON)]

par

loop [for each pedestrian UE in the camera area set]

19. POST /api/v1/sms/send [gpsi, text]

20. 200 OK

21. PUBLISH {topicBase}/{stationId}/DENM
[header, denm (management (eventPosition, relevanceDistance,
validityDuration), situation (eventType = accident))]

22. Relay the DENM to the vehicles subscribed to the event area